Exercise 2
Aggregate functions & Grouping

① SELECT COUNT id AS total numbe of employees
FROM employees_db;

total number of employees
10

② SELECT SUM(salary) AS total_salary_IT
FROM employees_db
WHERE department = 'IT';

total_salary_IT
220 000

③ SELECT AVG(salary) AS average_salary_HR
FROM employees_db
WHERE department = 'HR';

average_salary_HR
49 500

④ SELECT MIN(salary) AS min_salary
SELECT MAX(salary) AS max_salary
FROM employees_db;

min_salary	max_salary
48 000	62 000

⑤ SELECT department,
SUM(salary) AS total_salary_dept
FROM employees_db
GROUP BY department;

Department	
IT	220 000
HR	49 000
FINANCE	119 000
MARKETING	105 000

⑥ SELECT city,
COUNT * AS employes_per_city
FROM employees_db
Group By city;

city	employees_per_city
NY	2
Chicago	3
LA	2
San Fr	2
Houston	1

⑦ SELECT AVG (salary) AS average salary_dept
FROM employees_db
ORDER BY average salary_dept DESC
GROUP BY department;

department	average_salary_dept
IT	55 000
HR	49 500
Finance	59 500
Marketing	52 500

department	avg_salary_dept
Finance	59 500
IT	55 000
Marketing	52 500
HR	49 500

⑧ SELECT department,
SUM (salary) AS Total_salary
FROM employees_db
GROUP BY department
HAVING SUM (salary) > 100 000;

department	Total_salary
IT	220 000
FINANCE	119 000
MARKETING	105 000

⑨ SELECT city,
COUNT * AS total_employe_city
FROM employee_db
GROUP BY city
HAVING COUNT (*) > 1
ORDER BY total_employe_city DESC;

City	Total_employee_city
Chicago	3
NY	2
LA	2
SF	2

City	total employees
NY	2
Chicago	3
LA	2
S.F	2

(10)
```
SELECT department,
       AVG(salary) AS highest_avg_salary
FROM employees_db
GROUP BY department
ORDER BY highest_avg_salary
LIMIT 1;
```

department	highest_avg salary
Finance	501 500